



HudHudScript

What is HudHudScript

Agentic programming language with Loop Engineering, SOP, OOP and multilingual support.

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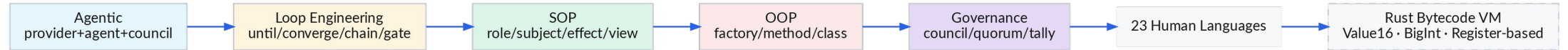
In one sentence

- HudHudScript is an agentic programming language where AI providers, agents, loop-engineering, SOP subjects, OOP objects, and governance councils are first-class constructs — written in any of 23 human languages, running on a Rust bytecode VM.

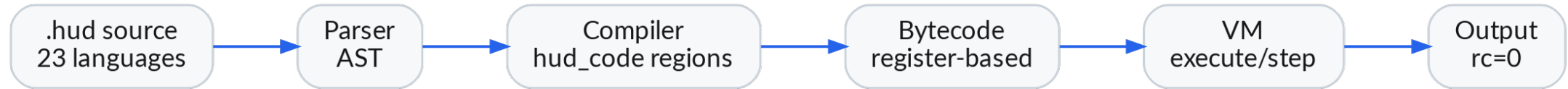
In one page

- First-class: provider/agent/council/role/subject/effect/loop/until/chain/gate
- Agentic: AI providers (ollama, OpenAI, DeepSeek) + agents with live LLM calls
- Loop Engineering: converge, retry, escalate, chain — engineered iteration
- SOP: subject-oriented programming with roles, effects, views
- OOP: factory/method, objects, classes
- Governance: councils with quorum, tally, human-in-the-loop gates
- Multilingual: 23 human languages in source code
- VM: register-based, Value16, BigInt, Rust bytecode

Paradigm stack



Architecture pipeline

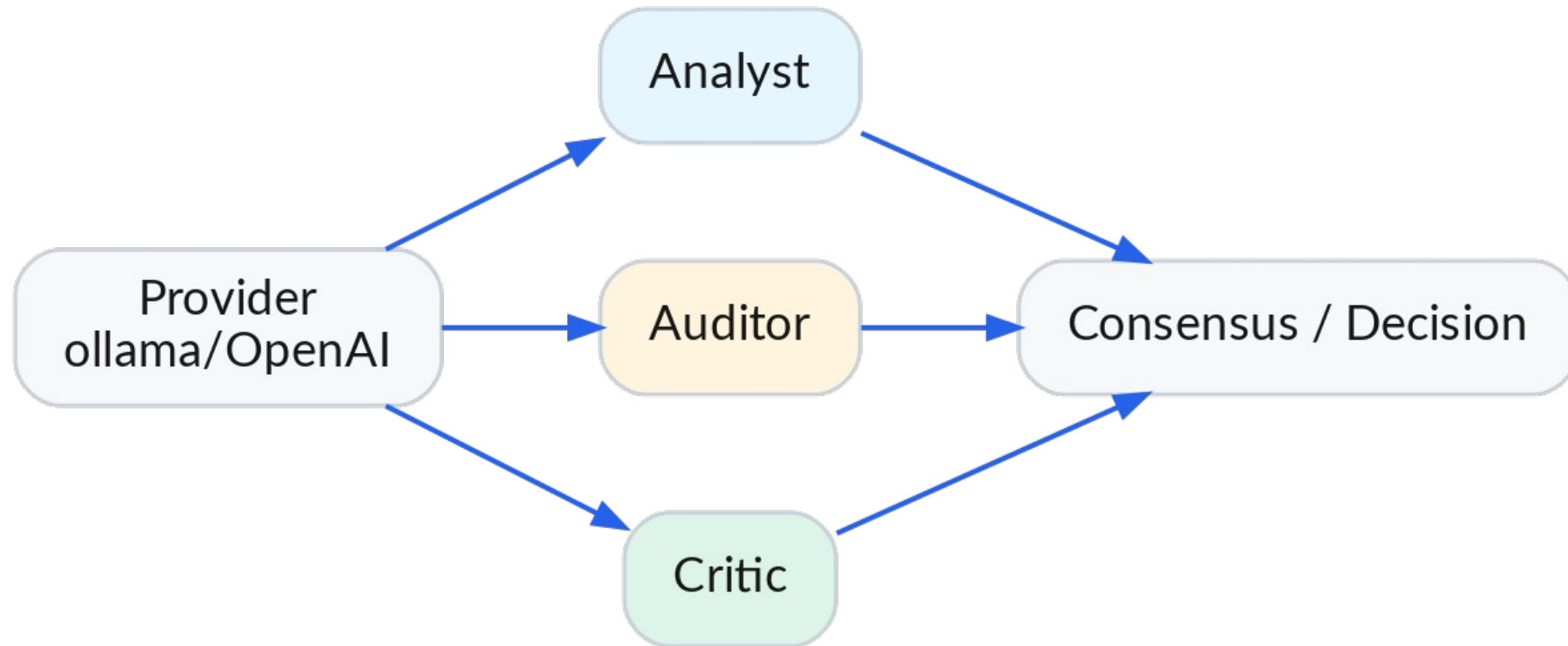


Agentic: live AI

```
agent OSINT_Analyst {  
  provider: "OSINT_Provider",  
  model: "gemma3:4b",  
  role: "Rate signal LOW/MEDIUM/HIGH risk. Score 1-10, 1-sentence reason."  
}  
agent Civil_Liberties_Auditor {  
  provider: "OSINT_Provider",  
  model: "gemma3:4b",  
  role: "If assessment targets individuals or violates privacy, reply VETO. Else APPROVE."  
}
```

LIVE ollama (192.168.1.7/gemma3:4b). Agent definitions for multi-agent systems.

Agent orchestration

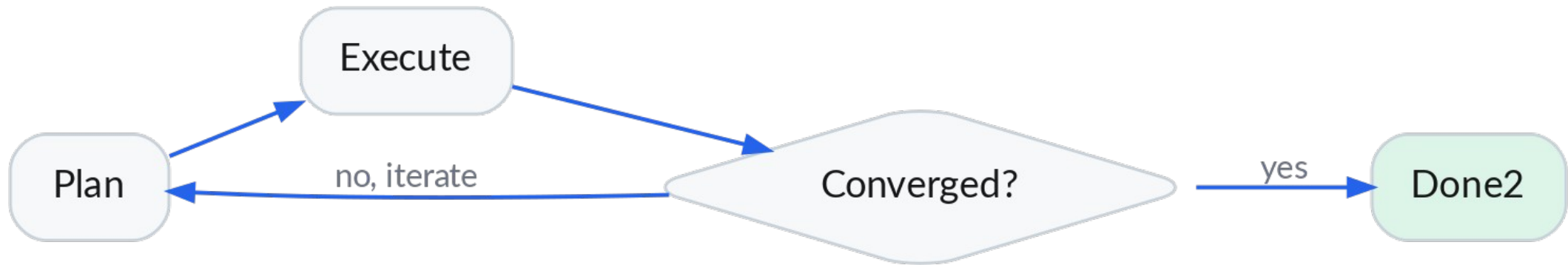


Loop Engineering

```
loop Publish {  
  step publish_artifact {  
    let release = Temp.dir("hudhud-publish-");  
    let artifact = release.path + "/calculator.js";  
    file.write(artifact, "console.log(2 + 3);\n");  
    result.output = exec.output("node " + artifact);  
    result.artifact = artifact;  
    print("Published: " + result.artifact);  
    gate publish_gate {  
      when result.output.length == 2  
        && file.exists(result.artifact) -> done  
      else -> fail  
    }  
  }  
}
```

Greedy allocation with while loop. Converges to optimal solution.

Loop-chain pattern

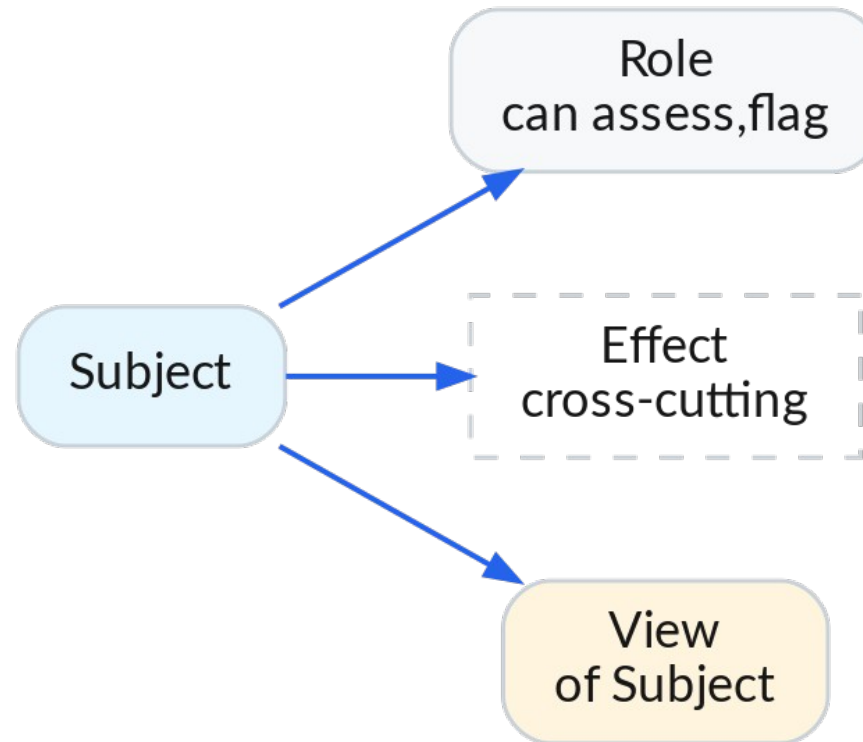


SOP: subject-oriented

```
subject ThreatCase has Analyst {  
  state risk: 0,  
  state stage: "new",  
  state approved: 0  
  on assess(self, intel_score) {  
    self.risk = intel_score  
    self.stage = "assessed"  
    print(" [ASSESS] risk=" + self.risk + " stage=" + self.stage)  
  }  
  ...  
}
```

SOP subjects with roles, state, and abilities. SOP warning 0.

SOP subject interaction



OOP: factory/method

```
trait Describable {  
  function describe();  
}  
class Account implements Describable {  
  private let balance = 0;  
  protected let owner = "";  
  constructor(owner, opening_balance) {  
    this.owner = owner;  
    this.balance = opening_balance;  
  }  
  private normalize(amount) {  
    if (amount < 0) { return 0; }  
    return amount;  
  }  
  ...  
}
```

Governance: real tally

```
law RequireEvidence {
  description: "A release decision must include test evidence";
  enforcement: mandatory;
  rules: ["tests-pass", "three-votes-required"];
}
constitution TrustedDelivery {
  description: "Policy for governed software delivery";
  laws: [
    { name: "ProtectSecrets", enforcement: "mandatory" },
    { name: "AuthorizeEffects", enforcement: "mandatory" }
  ];
}
rule ReleaseQuorum {
  description: "At least three of four reviewers must approve";
  action: "release";
}
```

AI Native & Multilingual (23 human languages)

- Source code in Turkish, Arabic, Japanese, Korean, Farsi, Hindi, and more.
- Keywords, identifiers, and comments in native script.
- One VM, many human languages — same bytecode.
- One programming language, many paradigms
- Developed for the age of agentic AI

AI Native keywords & development



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Website: hudhudscript.com GitHub: github.com/HudHudMind/hudhudscript

Patreon: patreon.com/cw/hudhudscript X: x.com/hudhud_script

Discord: discord.gg/UxEJ5MfH LinkedIn: linkedin.com/groups/27050016

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